Numerical Analysis

Chapter 6: Numerical Differentiation and Integration

Part 1, Numerical Differentiation

Department of Mathematics & Statistics
Missouri University of Science & Technology

Textbook: Classical and Modern Numerical Analysis: Theory, Methods and Practice
A. S. Ackleh, E. J. Allen, R. B. Kearfott and P. Seshaiyer
CRC Press, Boca Raton, FL, 2010

Outline

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Derivation with Taylor's theorem

Recall

Theorem

Suppose that $f(x)$ is a $(n+1)^{th}$ differentiable function on $[a,b]$ and $x_0 \in [a,b]$. Then for any $x \in [a,b]$, we have the following Taylor's expansion of $f(x)$ at x_0:

$$f(x) = P_n(x) + R_n(x),$$

where

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k \\ &= f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \cdots \\ &\quad + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n, \end{aligned}$$

Derivation with Taylor's theorem

Theorem (Continued)

$$R_n = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x - x_0)^{n+1}$$

for some $\xi \in [x_0, x]$ *(Lagrange form of the remainder)* ,

or

$$R_n = \frac{1}{n!} \int_{x_0}^{x} f^{(n+1)}(s)(x - s)^n \, ds$$

(Integral form of the remainder) .

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- Pick $n = 3$ in the Taylor's expansion:

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 \\ &\quad + \frac{1}{6} f'''(x_0)(x - x_0)^3 + \frac{1}{24} f^{(4)}(\xi)(x - x_0)^4. \end{aligned}$$

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- Replace x by $x + h$ and x_0 by x:

$$f(x + h) = f(x) + f'(x)h + \frac{1}{2} f''(x)h^2 + \frac{1}{6} f'''(x)h^3 + O\left(h^4\right).$$

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- We first consider the discretization of the first derivative $f'(x)$. Then

$$\begin{aligned} f'(x) &= \frac{f(x + h) - f(x)}{h} - \frac{1}{2} f''(x)h - \frac{1}{6} f'''(x)h^2 - O\left(h^3\right) \\ &= \frac{f(x + h) - f(x)}{h} + O\left(h\right). \end{aligned}$$

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- The mesh nodes are $t_j = a + jh,\ j = 0, 1, \cdots, J$.
- Then

$$\begin{aligned} f'(t_j) &= \frac{f(t_j + h) - f(t_j)}{h} + O(h) \\ &= \frac{f(t_{j+1}) - f(t_j)}{h} + O(h) \\ &\approx \frac{f_{j+1} - f_j}{h},\ j = 0, 1, \cdots, J - 1. \end{aligned}$$

Here f_j is the approximation of $f(t_j)$. This is called forward difference.

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- Recall the Taylor's expansion with $n = 3$:

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 \\ &\quad + \frac{1}{6} f'''(x_0)(x - x_0)^3 + \frac{1}{24} f^{(4)}(\xi)(x - x_0)^4. \end{aligned}$$

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- Replace x by $x - h$ and x_0 by x:

$$f(x - h) = f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 - \frac{1}{6}f'''(x)h^3 + O\left(h^4\right).$$

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$$\begin{aligned} f'(x) &= \frac{f(x) - f(x - h)}{h} + \frac{1}{2}f''(x)h - \frac{1}{6}f'''(x)h^2 + O\left(h^3\right) \\ &= \frac{f(x) - f(x - h)}{h} + O\left(h\right). \end{aligned}$$

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$$\begin{aligned} f'(t_j) &= \frac{f(t_j) - f(t_j - h)}{h} + O(h) \\ &= \frac{f(t_j) - f(t_{j-1})}{h} + O(h) \\ &\approx \frac{f_j - f_{j-1}}{h}, \ j = 1, \cdots, J. \end{aligned}$$

This is called backward difference.

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- Recall

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(x)h^3 + O\left(h^4\right),$$

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$$f(x-h) = f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 - \frac{1}{6}f'''(x)h^3 + O\left(h^4\right).$$

- Subtract the second equation from the first one:

$$f(x+h) - f(x-h) = 2f'(x)h + \frac{1}{3}f'''(x)h^3 + O\left(h^4\right).$$

Derivation with Taylor's theorem

- Then

$$\begin{aligned} f'(x) &= \frac{f(x+h)-f(x-h)}{2h} - \frac{1}{12}f'''(x)h^2 + O\left(h^3\right) \\ &= \frac{f(x+h)-f(x-h)}{2h} + O\left(h^2\right). \end{aligned}$$

This is second order!

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- Hence

$$\begin{aligned} f'(t_j) &= \frac{f(t_j+h)-f(t_j-h)}{2h} + O\left(h^2\right) \\ &= \frac{f(t_{j+1})-f(t_{j-1})}{2h} + O\left(h^2\right) \\ &\approx \frac{f_{j+1}-f_{j-1}}{2h},\ j=1,\cdots,J-1. \end{aligned}$$

This is called centered difference.

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- Now let's turn to the discretization of the second derivative $f''(x)$.

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- Pick $n = 4$ in the Taylor's expansion:

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \\ &\quad \frac{1}{6}f'''(x_0)(x - x_0)^3 + \frac{1}{24}f^{(4)}(x_0)(x - x_0)^4 \\ &\quad + \frac{1}{120}f^{(5)}(\xi)(x - x_0)^5. \end{aligned}$$

Derivation with Taylor's theorem

- Replace x by $x+h$ and x_0 by x:

$$\begin{aligned} f(x+h) &= f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(x)h^3 \\ &\quad + \frac{1}{24}f^{(4)}(x)h^4 + O\left(h^5\right). \end{aligned}$$

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- Replace x by $x-h$ and x_0 by x:

$$\begin{aligned} f(x-h) &= f(x)-f'(x)h+\frac{1}{2}f''(x)h^2-\frac{1}{6}f'''(x)h^3 \\ &\quad +\frac{1}{24}f^{(4)}(x)h^4+O\left(h^5\right). \end{aligned}$$

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- Add the above two equations together:

$$f(x+h)+f(x-h)=2f(x)+f''(x)h^2+\frac{f^{(4)}(x)}{12}h^4+O\left(h^5\right) \quad .$$

Derivation with Taylor's theorem

- Then

$$\begin{aligned} f''(x) &= \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \frac{f^{(4)}(x)}{12} h^2 + O\left(h^3\right) \\ &= \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O\left(h^2\right). \end{aligned}$$

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This is called centered second difference.

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Derivation via Lagrange polynomial representation

Recall the following theorem from section 4.3:

Theorem

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Derivation via Lagrange polynomial representation

Recall the following definition and theorem from section 4.3:

Definition

$p(x) = \sum_{k=0}^{n} f(x_k)L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

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Theorem (II)

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$.

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- For $m = 1$: $f'(x) \approx p'(x) = \sum_{k=0}^{n} f(x_k) L'_k(x)$.
- And $f'(x_l) \approx p'(x_l) = \sum_{k=0}^{n} f(x_k) L'_k(x_l)$.

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- Basic idea: $f^{(m)}(x) \approx p^{(m)}(x)$ where $p(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$.
- For $m = 1$: $f'(x) \approx p'(x) = \sum_{k=0}^{n} f(x_k) L'_k(x)$.
- And $f'(x_l) \approx p'(x_l) = \sum_{k=0}^{n} f(x_k) L'_k(x_l)$.
- Don't forget the remainder: $\frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$ where $W(x) = \prod_{i=0}^{n}(x - x_i)$.
- So

$$\begin{aligned} f(x) &= p(x) + \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!} \\ &= \sum_{k=0}^{n} f(x_k) L_k(x) + \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}. \end{aligned}$$

Derivation via Lagrange polynomial representation

- $f'(x) = \sum_{k=0}^{n} f(x_k) L'_k(x) + W'(x) \frac{f^{(n+1)}(\xi(x))}{(n+1)!} + \frac{d}{dx} \frac{f^{(n+1)}(\xi(x))}{(n+1)!} W(x)$.

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- Recall $W(x) = \prod_{i=0}^{n}(x - x_i)$.
- So $W'(x) = \sum_{j=0}^{n} \prod_{i=0, i\neq j}^{n}(x - x_i) =$

  $\prod_{i=1}^{n}(x - x_i) + \prod_{i=0, i\neq 1}^{n}(x - x_i) + \cdots + \prod_{i=0}^{n-1}(x - x_i)$.

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- And $W'(x_l) = \prod_{i=0, i\neq l}^{n}(x_l - x_i)$ for any $0 \leq l \leq n$.
- Also, $W(x_l) = \prod_{i=0}^{n}(x_l - x_i) = 0$ for any $0 \leq l \leq n$.

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- Also, $W(x_l) = \prod_{i=0}^{n}(x_l - x_i) = 0$ for any $0 \leq l \leq n$.
- Hence $f'(x_l) = \sum_{k=0}^{n} f(x_k)L'_k(x_l) + \prod_{i=0, i\neq l}^{n}(x_l - x_i)\frac{f^{(n+1)}(\xi(x_l))}{(n+1)!}$.

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Derivation via Lagrange polynomial representation

- Example 1: Pick $n = 2$, $x_1 = x_0 + h$, and $x_2 = x_1 + h$.
- Then $f'(x_l) = \sum_{k=0}^{2} f(x_k) L_k'(x_l) + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6}$.
- Recall

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

- Then

$$\begin{aligned} L_0(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}, \\ L_1(x) &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}, \\ L_2(x) &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}. \end{aligned}$$

Derivation via Lagrange polynomial representation

- Hence

$$L_0'(x) = \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)},$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)},
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
L_2'(x) &= \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
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L_2'(x) &= \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}.
\end{aligned}
$$

- And

$$
L_0'(x_l) = \frac{2x_l - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)},
$$

Derivation via Lagrange polynomial representation

- Hence

$$
\begin{aligned}
L_0'(x) &= \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}, \\
L_1'(x) &= \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}, \\
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\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Hence for $l = 0, 1, 2$:

$$f'(x_l) = \sum_{k=0}^{2} f(x_k) L'_k(x_l) + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6}$$

Derivation via Lagrange polynomial representation

- Hence for $l = 0, 1, 2$:

$$
\begin{aligned}
f'(x_l) &= \sum_{k=0}^{2} f(x_k) L_k'(x_l) + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6} \\
&= f(x_0) \frac{2x_l - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1) \frac{2x_l - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\
&\quad + f(x_2) \frac{2x_l - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq l}^{2} (x_l - x_i) \frac{f'''(\xi(x_l))}{6}.
\end{aligned}
$$

Derivation via Lagrange polynomial representation

- Pick $l = 1$:

$$\begin{aligned} f'(x_1) &= f(x_0)\frac{2x_1 - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1)\frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2)\frac{2x_1 - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \prod_{i=0, i\neq 1}^{2}(x_1 - x_i)\frac{f'''(\xi(x_1))}{6} \end{aligned}$$

Derivation via Lagrange polynomial representation

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Derivation via Lagrange polynomial representation

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$$
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&= -\frac{1}{2h}f(x_0) + \frac{1}{2h}f(x_2) - \frac{f'''(\xi(x_1))}{6}h^2 \\
&= \frac{1}{2h}[f(x_1 + h) - f(x_1 - h)] - \frac{f'''(\xi(x_1))}{6}h^2.
\end{aligned}
$$

Derivation via Lagrange polynomial representation

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- Choose $x_1 = t_j$, then we get the centered difference:
  $f'(t_j) = \frac{1}{2h}[f(t_j + h) - f(t_j - h)] - \frac{f'''(\xi(t_j))}{6}h^2.$

Roundoff error

- Numerical differentiation: use the values of a function at mesh nodes to approximate the derivative values of the function.

Roundoff error

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- What if the values of a function at mesh nodes can not be expressed exactly due to roundoff?

Roundoff error

- Numerical differentiation: use the values of a function at mesh nodes to approximate the derivative values of the function.
- What if the values of a function at mesh nodes can not be expressed exactly due to roundoff?
- Conclusion: the total error caused by a small roundoff error could be HUGE!

Roundoff error

- Example 2: Consider the forward difference

$$f'(x_j) \approx \frac{f_{j+1} - f_j}{h} \left(= \frac{f(x_{j+1}) - f(x_j)}{h} \right).$$

Let $\tilde{f}_{j+1}$ and $\tilde{f}_j$ be the numbers stored in the computer memory for f_{j+1} and f_j respectively. Assume $f_{j+1} = \tilde{f}_{j+1} + e(x_{j+1})$ and $f_j = \tilde{f}_j + e(x_j)$ where $e(x_{j+1})$ and $e(x_j)$ are roundoff errors. In general $|e(x)| \leq \varepsilon |f(x)|$ for some relative error ε. Estimate the total error

$$\left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right|.$$

Roundoff error

- The forward difference scheme comes from the expansion

$$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h)$$

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$$\Rightarrow \quad f'(x_j) = \frac{f(x_j+h) - f(x_j)}{h} + O(h)$$

$$\Rightarrow \quad f'(x_j) = \frac{f(x_{j+1}) - f(x_j)}{h} + O(h)$$

Roundoff error

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Roundoff error

- Then

$$\left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right|$$

Roundoff error

- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \end{aligned}$$

Roundoff error

- Then

$$\begin{aligned} & \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\ \leq & O(h) + \left| \frac{(f_{j+1} - \tilde{f}_{j+1}) - (f_j - \tilde{f}_j)}{h} \right| \end{aligned}$$

Roundoff error

- Then

$$
\begin{aligned}
& \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & O(h) + \left| \frac{(f_{j+1} - \tilde{f}_{j+1}) - (f_j - \tilde{f}_j)}{h} \right| \\
\leq & O(h) + \left| \frac{e(x_{j+1}) - e(x_j)}{h} \right|
\end{aligned}
$$

Roundoff error

- Then

$$
\begin{aligned}
& \left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & \left| f'(x_j) - \frac{f_{j+1} - f_j}{h} \right| + \left| \frac{f_{j+1} - f_j}{h} - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right| \\
\leq & O(h) + \left| \frac{(f_{j+1} - \tilde{f}_{j+1}) - (f_j - \tilde{f}_j)}{h} \right| \\
\leq & O(h) + \left| \frac{e(x_{j+1}) - e(x_j)}{h} \right| \\
\leq & O(h) + \frac{|e(x_{j+1})| + |e(x_j)|}{h}.
\end{aligned}
$$

Roundoff error

- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
- Then

$$\left| f'(x_j) - \frac{\tilde{f}_{j+1} - \tilde{f}_j}{h} \right|$$

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- Recall that $|e(x)| \leq \varepsilon |f(x)|$.
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- Then

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- The error could be HUGE if h is much smaller than ε!

Roundoff error

- Example 2: Consider the forward difference for $f(x) = ln(x)$:

$$ln'(x) \approx \frac{ln(x+h) - ln(x)}{h}.$$

Compute the numerical differentiation for $x = 3$ and $h = 10^{-n}$ $(n = 1, 2, \cdots, 16)$. Use $ln'(x) = \frac{1}{x}$ to compute the errors $\left| ln'(3) - \frac{ln(3+h)-ln(3)}{h} \right|$ of the numerical solutions.

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- Open your Matlab to code for this simple problem!

Roundoff error

- Numerical results:

h	numerical differentiation	error
10^{-1}	0.327898228229910	0.005435105103423
10^{-2}	0.332779009267470	0.000554324065863
10^{-3}	0.333277790120601	0.000055543212733
10^{-4}	0.333327777903847	0.000005555429486
10^{-5}	0.333332777802120	0.000000555531213
10^{-6}	0.333333277868775	0.000000055464559
10^{-7}	0.333333329827212	0.000000003506121
10^{-8}	0.333333338708997	0.000000000375663
10^{-9}	0.333333360913457	0.000000027580124
10^{-10}	0.333333360913457	0.000000027580124
10^{-11}	0.333333360913457	0.000000027580124
10^{-12}	0.333510996597397	0.000177663264064
10^{-13}	0.333066907387547	0.000266425945786
10^{-14}	0.355271367880050	0.021938034546717
10^{-15}	0.444089209850063	0.110755876516729
10^{-16}	0	0.333333333333333

Topics in numerical differentiation: Independent study problems

- Higher order difference schemes for $f'(x)$ by using more nodes.

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- .....